development. During organogenesis the various organs of the foetus are established from the different germ layers attaining a functional state.

Formation of Placenta

The placenta is a disc shaped structure attached to the uterine wall and is a temporary association between the developing embryo and maternal tissues. It allows the exchange of food materials, diffusion of oxygen, excretion of nitrogenous wastes and elimination of carbon dioxide. A cord containing blood vessels that connects the placenta with the foetus is called the **umbilical cord**.

Pregnancy (Gestation)

It is the time period during which the embryo attains its development in the uterus. Normally gestation period of human last for about 280 days. During pregnancy the uterus expands upto 500 times of its normal size.

Parturition (Child Birth)

Parturition is the **expulsion** of young one from the mother's uterus at the end of gestation. **Oxytocin** from the posterior pituitary stimulates the **uterine contractions** and provides force to expel the baby from the uterus, causing birth.



Sometimes ovaries releases two eggs and each is fertilised by a different sperm, resulting in **Non-Identical Twins (Fraternal Twins)**. If single egg is fertilised and then divides into two foetus, **Identical Twins** develop.

Lactation

The process of milk production after child birth from mammary glands of the mother is called lactation. The first fluid which is released from the mammary gland after child birth is called as **colostrum.** Milk production from alveoli of mammary glands is stimulated by **prolactin** secreted from the anterior pituitary. The **ejection of milk** is stimulated by posterior pituitary hormone **oxytocin**.

Info bits

The milk produced from the breast during the first 2 to 3 days after child birth is called colostrum. It contains immune substances and provides immunity to the new born which is essential for the body.

17.11. Reproductive Health

According to World Health Organization (WHO) reproductive health means a total well being in all aspects of reproduction, ability to reproduce and regulate fertility, women's ability to undergo pregnancy and safe child birth,maternal and infant survival and well being.

Several measures were undertaken by the government to improve the reproductive health of the people by launching **National Health Programmes** such as the

i. Family Welfare Programme
ii. Reproductive and Child Health Care (RCH) Programme

Family welfare programme: The National Family Welfare Programme is a comprehensive scheme which includes:

1. Maternal and child health care (MCH)
2. Immunization of mothers, infants and children.
3. Nutritional supplement to pregnant women and children.
4. Contraception with health education, to motivate couples to accept contraceptive methods and to have small family norms, which improve economic status, living status and the quality of life.

Reproductive and Child Health Care (RCH) Programme: It has integrated all services which include

- Pregnancy and child birth
- Postnatal care of the mother and child

10th_Science Unit-17.indd 253 11-01-2020 15:15:43